

Zomato Restaurant EDA Analysis Report

Business Problem

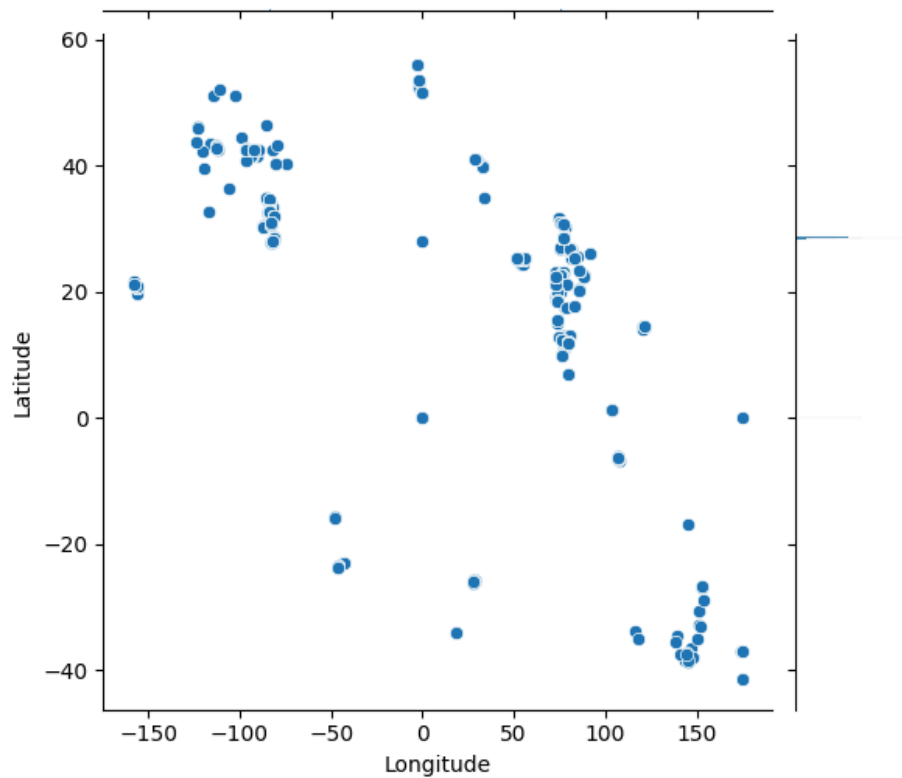
Restaurant ratings are influenced by country, delivery options, pricing, and table booking. The goal is to explore Zomato data, find patterns in ratings and costs, and build a model to predict aggregate restaurant ratings.

Hypothesis

Restaurants with online delivery have higher engagement. Higher-priced restaurants tend to score better. India dominates the dataset. Most restaurants remain unrated. Table booking correlates with premium categories.

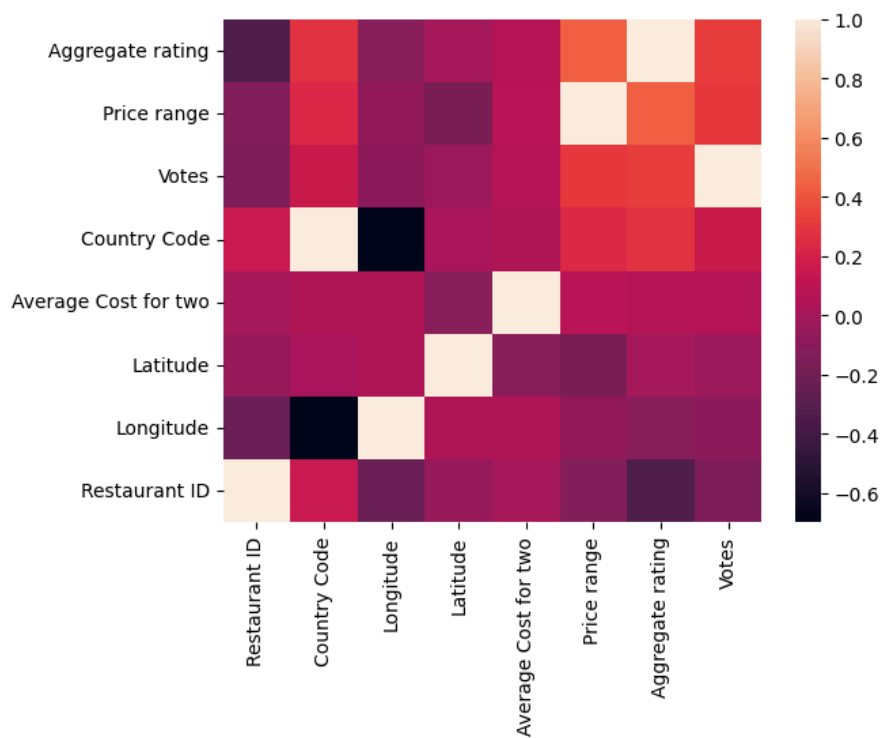
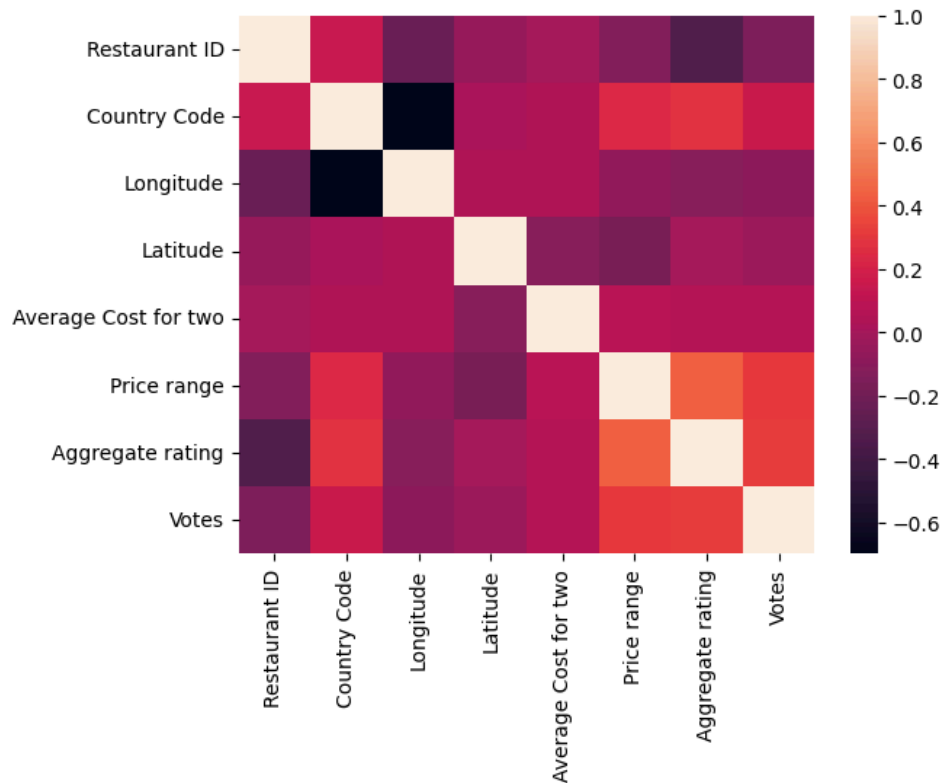
Geo Distribution — Longitude vs Latitude

The joint plot shows that most restaurant entries are clustered in India, with sparse presence across other continents.



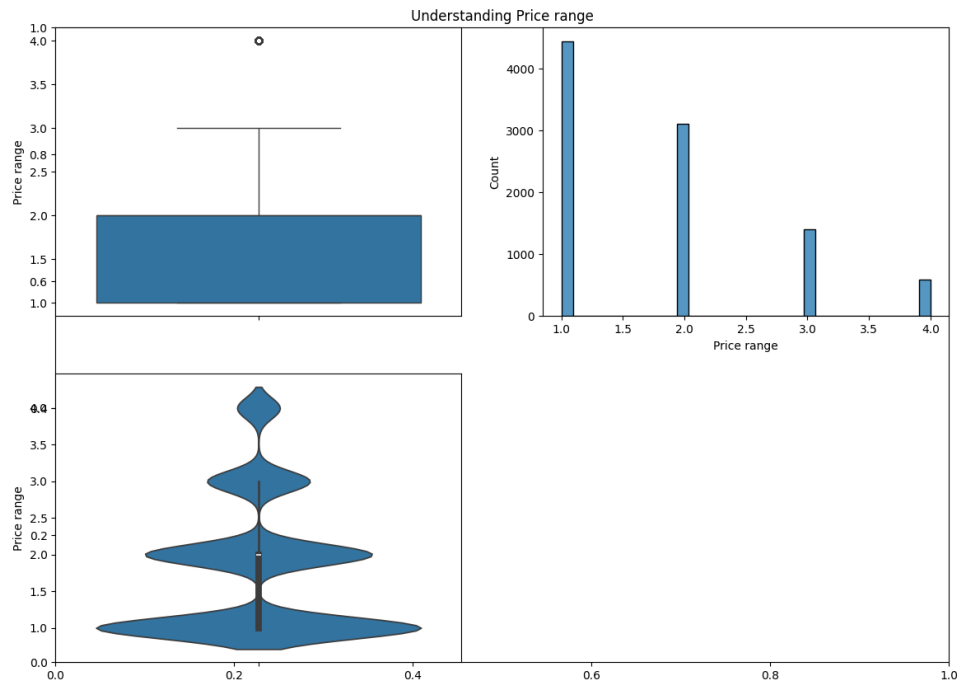
Correlation Heatmaps

Price range shows the strongest positive correlation with Aggregate rating. Votes also align positively with ratings. The second image is shown as the 'Aggregate Rating' arranged in Ascending order



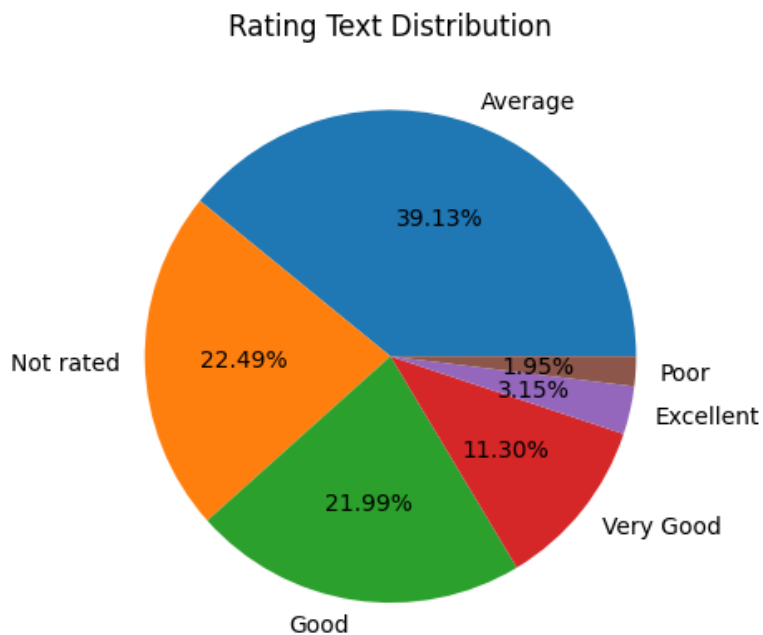
Univariate Analysis — Price Range

Most restaurants fall in price ranges 1 and 2. Very few sit in range 4, indicating budget-friendly dining dominates the dataset.



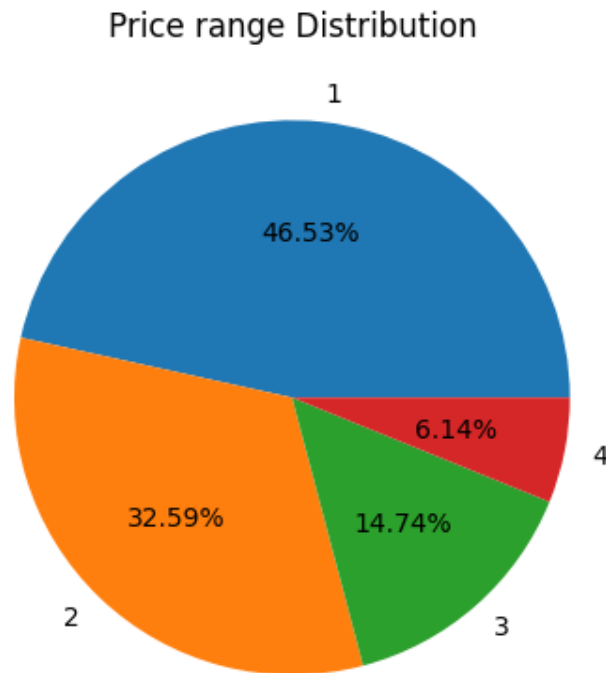
Rating Text Distribution

Over 44% of restaurants were never rated. Among rated ones, 'Good' and 'Very Good' lead. Excellent ratings are rare at ~4%.



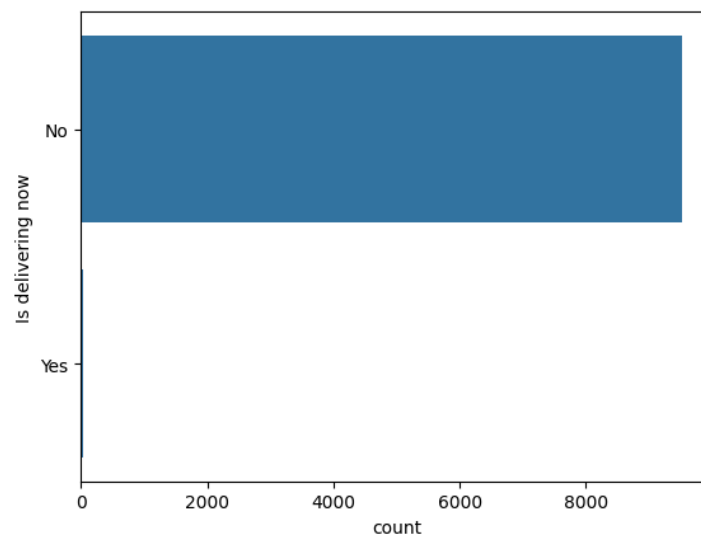
Price Range Distribution

Price range 1 dominates with over half the share. Range 4 is the rarest, confirming that premium dining is underrepresented.



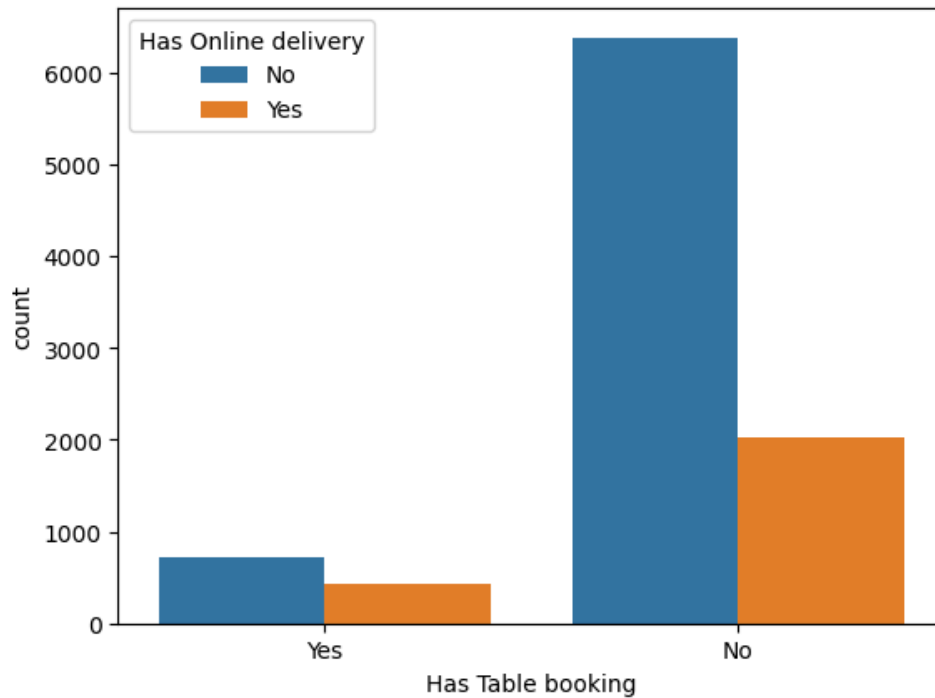
Is Delivering Now

A large majority of restaurants were not actively delivering at the time of data collection, reflecting the early stage of online food delivery adoption.



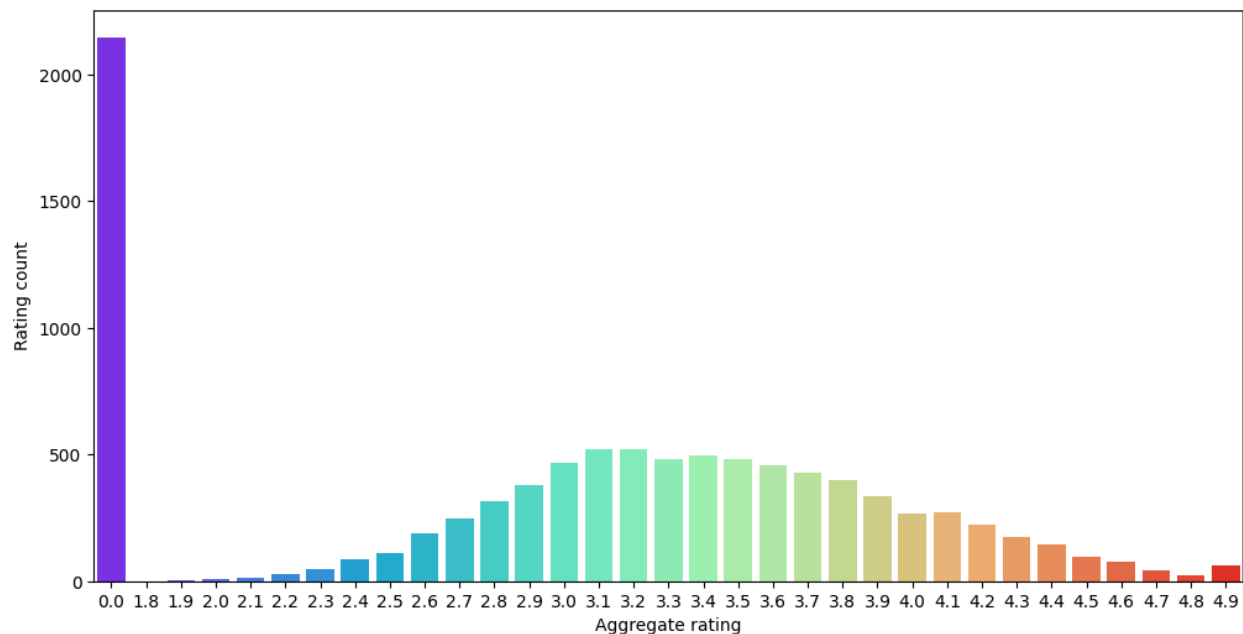
Multivariate — Table Booking vs Online Delivery

Restaurants with table booking are more likely to also offer online delivery. Most restaurants in the dataset offer neither.



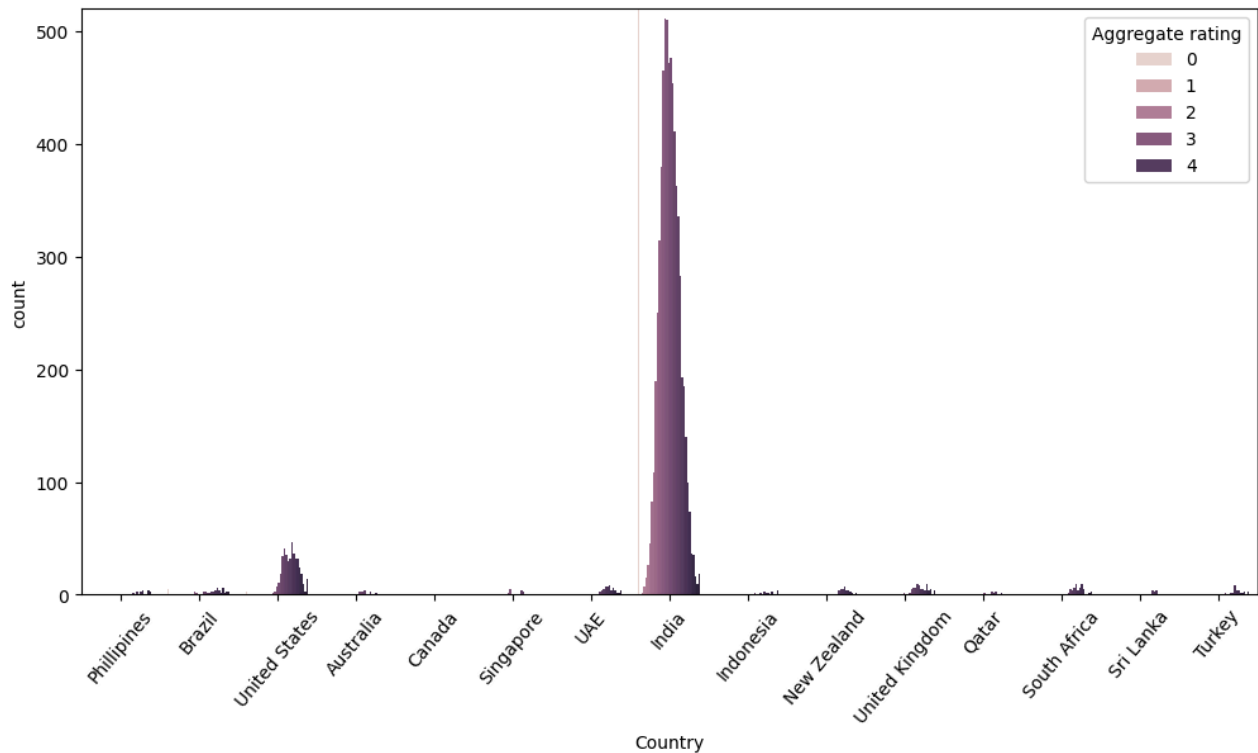
Rating Count vs Aggregate Rating

Ratings of 0.0 have the highest count — driven by unrated restaurants, mostly from India. Scores between 3.5 and 4.5 are most common among actually rated entries.



Country vs Aggregate Rating

India shows the highest unrated count due to dataset dominance. USA and UK have a more balanced rating spread across score buckets.



Model Training and Results

```
rf_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('regressor', RandomForestRegressor(random_state=42))
])
```

R2 Score - Linear Regression : 92.34%

R2 Score - Random Forest : 98.75%

Iteration log:

```
# Without Ordinal Mapping : 79.72%
# After Ordinal Mapping : 91.87%
# + Longitude, Latitude : 92.19%
# + Restaurant ID : 92.28%
# + City : 92.34%
# Random Forest : 98.75%
```

Ordinal encoding of Rating text alone boosted R^2 from 79.72% to 91.87%. Random Forest ultimately achieved 98.75% R^2 .

Data-Driven Insights

- A huge chunk of restaurants were never rated — especially in India.
- Only a few restaurants allow table booking, usually the fancier ones.
- Online delivery was not yet common at the time of data collection.
- The majority fall in lower price ranges (1 or 2 out of 4).
- 9 missing cuisine values were filled with 'Fast Food' — no dominant pattern found per city.

Suggestions

1. Promote well-rated but low-visibility restaurants to improve platform diversity.
2. Encourage unrated restaurants to collect reviews via targeted nudges.
3. Apply dynamic pricing insights on high-demand areas to maximize revenue.
4. Use cuisine-level and booking behavior features for further model improvement.

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